

Multiple-host fungi are the most frequent and abundant ectomycorrhizal types in a mixed stand of Douglas fir (*Pseudotsuga menziesii*) and bishop pine (*Pinus muricata*)

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SUMMARY

The ectomycorrhizal fungal associations of Douglas fir (*Pseudotsuga menziesii* D. Don) and bishop pine (*Pinus muricata* D. Don) were investigated in a mixed forest stand. We identified fungi directly from field-collected ectomycorrhizal (ECM) root tips using PCR-based methods. Sixteen species of fungi were found, of which twelve associated with both hosts. *Rhizopogon parksii* Smith was specific to Douglas fir. Three other species colonized only one of the hosts, but were too infrequent to draw conclusions about specificity. Seventy-four percent of the biomass of ECM root tips sampled in the stand were colonized by members of the Thelephoraceae and Russulaceae. All 12 species of fungi that associated with both tree species did so within a 10 × 40 cm soil volume, suggesting that individual fungal genotypes linked the putatively competing tree hosts.

Key words: Douglas fir (*Pseudotsuga menziesii* D. Don), bishop pine (*Pinus muricata* D. Don), ectomycorrhiza, specificity, ITS–RFLP.

INTRODUCTION

Harley & Smith (1983) reported that most ECM fungi can associate with multiple host species, giving them greater access to sources of carbon. However, Molina, Massicotte & Trappe (1992) predict that more ECM fungi will prove to have a narrow host range (associating with plants from a single genus) than currently appreciated, citing physiological adaptations to marginal habitats or plant chemicals as selection pressures toward specificity. Both of these views focus on the number and type of taxa that are host-specific or non-host-specific. There are currently no data published that assess whether host-specific or non-host-specific fungi dominate in the field. To understand how ECM fungi affect plant community dynamics, it is important to know (1) if fungi in a mixed plant community associate with multiple hosts or are partitioned between them and (2) whether host-specific or non-host-specific fungi are more abundant in terms of their ECM biomass.

Ectomycorrhizal fungi that associate with multiple hosts in a stand can translocate nutrients between hosts (Finlay & Read, 1986; Duddridge *et al.*, 1988; Söderström, Finlay & Read, 1988; Simard *et al.*, 1997b). This suggests that competition for soil nutrients by different plant species might be a function of their individual interactions with the common symbionts (Finlay, 1989). Conversely, host-specific fungi can provide exclusive access for different plant species to soil nutrients (Molina *et al.*, 1992), which suggests that competition for these nutrients between plants is in part assumed by the different fungal species.

Few studies document that fungal species associate with more than one host within a single forest stand. Trappe & Fogel (1977) found a single hypha of *Cenococcum geophilum* connected to ectomycorrhizas of Douglas fir (*Pseudotsuga menziesii*) and western hemlock (*Tsuga heterophylla*). Such observations are limited by the delicate and often cryptic nature of mycelia. Sporocarp collections in pure stands of different plants show that many fungi associate with multiple hosts but cannot predict whether host preferences might occur at the root level in mixed stands (Trappe, 1962; Bills, Holtzman & Miller,

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1986). In many cases laboratory bioassays (Perry *et al.*, 1989b; Massicotte *et al.*, 1994; Smith, Molina & Perry, 1995) and pure culture synthesis experiments (Zak 1976a, b; Molina & Trappe, 1982a, b; Molina & Trappe, 1994) support the assumption that sporocarp associations are good indicators of levels of specificity. However, ecological conditions might narrow host ranges observed in glasshouse experiments because of biotic and abiotic factors unique to the field (Harley & Smith, 1983; Molina *et al.*, 1992) and in any case, many common fungal taxa have not been successfully maintained in culture; direct field based studies of ECM associations in mixed stands are needed to demonstrate association patterns.

The primary objective of this work was to investigate if the fungal associates were different on the roots of Douglas fir (*Pseudotsuga menziesii*) and bishop pine (*Pinus muricata*) where roots of both trees intermingled. To avoid the problems discussed above we used molecular-based methods to identify fungal and plant species from ectomycorrhizas collected in the field. From these data we ascertained the relative abundance of fungal species and whether they occurred on one or both hosts.

MATERIALS AND METHODS

Site characteristics

The study site is located within the Point Reyes National Seashore along the coast of central California at 38° 2' 58" longitude, 22° 51' 1" latitude, 121 m elevation. It is across a ravine from a site that has been under investigation since 1991 (Gardes & Bruns, 1996). The local Mediterranean climate is moderated by summer fog. The average annual air temperature is 11–14 °C and the average rainfall typically ranges between 63 and 88 cm. Peak precipitation occurs between January through March, but fog drip provides moisture during the summer.

The portion of forest studied was *c.* 625 m² in size and consisted of a mixed stand of Douglas fir (*Pseudotsuga menziesii* D. Don) and bishop pine (*Pinus muricata* D. Don) in about a three to one ratio of stems, respectively. Stand composition changed from a relatively even percentage of Douglas fir and bishop pine at the south end, to dominance by Douglas fir at the north end (Fig. 1). East of the study site was almost pure, young Douglas fir and to the west the forest had larger bishop pine and Douglas fir mixed with an understory of arbuscular mycorrhizal shrubs including *Baccaris pilularis* DC, *Rhamnus californica* Eschsch. and *Rubus ursinus* Cham. & Schldl. In the study site, the understory was sparse and consisted of a few arbuscular mycorrhizal herbs. We collected ECM samples in January and February 1995. After completion of the study, a large wildfire burned the study site in October 1995, killing all of the trees. A previous fire

occurred approx. 35 yr ago as revealed by tree ring counts of several trees in the area.

ECM sampling and morphotyping

Our aim was to collect soil cores likely to contain ectomycorrhizas from both conifer species. Because the root distribution of the two tree species was unknown, we developed a sampling scheme that was biased towards locations where the two tree hosts occurred within 5 m of each other (Fig. 1). We located a site non-randomly so that at *c.* 5-m intervals, pairs of trees occurred in which one was Douglas fir and the other bishop pine. About 2–3 m separated the individuals of each tree pair. We took five soil cores distributed evenly between each tree pair from Douglas fir to bishop pine. Ectomycorrhizas were collected with a 10-cm diameter soil corer driven to a depth of 40 cm (3141 cm³ soil). Twenty-five cores were taken in total, a maximum given the time constraints for processing the samples. The sample size was relatively low and did not allow for much statistical analysis. However, the molecular tools gave us precise and novel information about the fungal community.

Soils were stored at 4 °C until processed. Soil cores were soaked in water for 24 h at room temperature before rinsing. Ectomycorrhizas were isolated by rinsing the soils through a no. 35 soil sieve (0.5-mm mesh size) before sorting by morphology under a dissecting microscope.

Criteria for sorting ectomycorrhizas included colour, features of the mantle such as cystidia, branching pattern and characteristics of rhizomorphs following Agerer (1987–1996). Douglas-fir and bishop-pine ectomycorrhizas were sorted based on their distinct branching morphologies (pinnate and bifurcate, respectively). The tree species of unbranched, single root tips were later identified using molecular methods. One ECM root tip from each morphological type (morphotype) was sectioned to confirm the presence of a Hartig net. ECM root samples were lyophilized in preparation for biomass measurements, molecular analyses and storage. All samples were processed to the lyophilization step within 2 wk after removal from the field. To quantify the ectomycorrhizas we measured the d. wt of each morphotype for each core and each tree species.

We intentionally split the ectomycorrhizas from each core into more categories than was necessary and combined like types only after molecular identifications were complete. In this way, we avoided combining different species with similar morphotypes. This was particularly important for morphotypes within a family group (Russulaceae, Thelephoraceae, etc.), which often had subtle differences that were easily recognized only after experience. This approach allowed for a rapid sorting

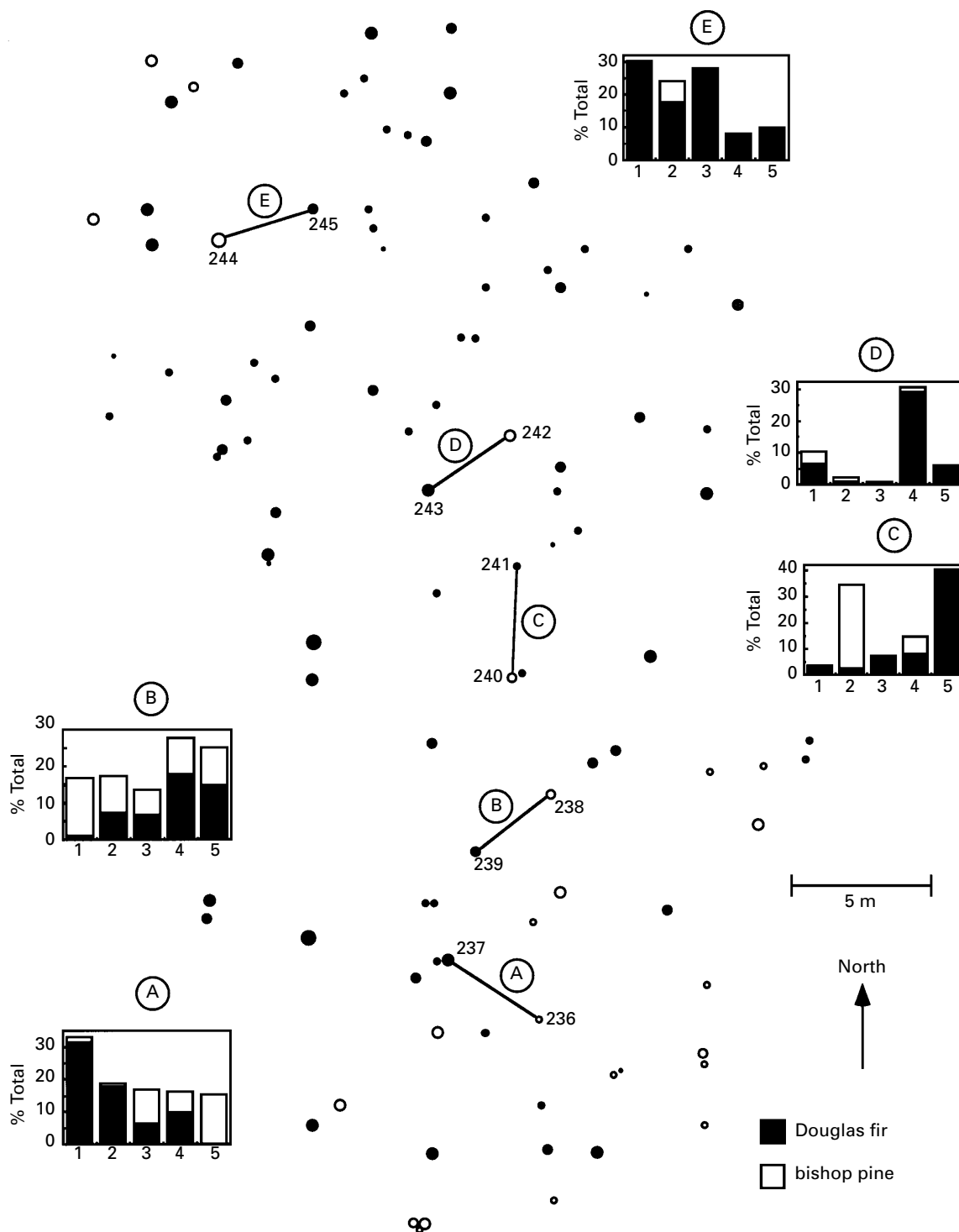


Figure 1. Map of the research site. Five cores were taken along the lines shown between each numbered tree pair; cores one to five were always taken from Douglas fir to bishop pine, respectively. Histograms show percentage of ectomycorrhizas by weight partitioned by tree species in each core. Circles represent tree species in the stand with the diameters and positions roughly to scale. Only trees within 10 m of the plots are shown.

of root tips so as to preserve DNA for the molecular identification of the symbionts.

Molecular identification of ECM symbionts

We used PCR-based methods to identify plant and fungal symbionts from most ectomycorrhizas to species or family group level. DNA extraction, PCR

amplifications and RFLP protocols followed Gardes & Bruns (1993). We identified fungi and plants from ectomycorrhizas by comparing RFLP patterns to those from voucher specimens of sporocarps or plant leaves; samples which displayed identical RFLP patterns, including any sub-molar bands, were considered to belong to the same RFLP type. Molecular identifications were repeated at least once

for each core/tree/fungus morphotype sample. Because of our splitting approach in sorting, and because most types occurred in several cores, the ectomycorrhizas were repeatedly sampled for molecular identification, giving us confidence that our final morphotypes represented single species. All root tips within a morphotype that consisted of individual, unbranched root tips were sampled for tree identification.

The plant-specific primer pair 28KJ and TW14 was used to amplify a portion of the 28S gene in the nuclear rRNA repeat from the ECM root extracts (Cullings, 1992). The two plant species can be unambiguously differentiated with RFLP patterns generated when this region is digested with the restriction enzyme *Mbo* I. Fungi were identified utilizing sequence differences in the internal transcribed spacer (ITS) of the nuclear rRNA repeat using either the fungal specific primer pair ITS1f and ITS4 (Gardes *et al.*, 1991) or the basidiomycete-specific primer pair ITS1f and ITS4b (Gardes & Bruns, 1993). Fungal ITS-RFLP patterns were produced using restriction enzymes *Hinf* I, *Alu* I and *Mbo* I. RFLP band sizes were estimated using GelReader® version 2.0.5 (National Center for Supercomputing Applications, University of Illinois, USA); this gives about a 5% (up to 10%) error in size estimates of ITS-RFLP bands of the same sample run on different gels.

If the fungus species remained unknown, we sequenced one of two regions for phylogenetic analysis. DNA sequences were produced with the cyclic reaction termination method using fluorescence-labelled dideoxynucleotide triphosphates and following the instructions provided with the PRISM® Ready Reaction DyeDeoxy® Terminator Cycle Sequencing Kit (Perkin-Elmer Corporation). Electrophoresis and data collection were done on an ABI Model 373A DNA Sequencer (Perkin-Elmer Corporation) to which a Macintosh Quadra® 650 computer was connected. DNA Sequencing Analysis (version 2.01) and SeqEd (version 1.03) were used to process raw data. Basidiomycete family placement was conducted using a working version of the mitochondrial large subunit rRNA gene database (Cullings, Szaro & Bruns, 1996; Gardes & Bruns, 1996; Taylor & Bruns, 1997; Bruns *et al.*, 1998). If necessary, the fungal division (i.e. Ascomycota vs. Basidiomycota) of an unknown was identified using a database of the 5.8S nuclear rRNA gene, located within the ITS region (Cullings & Vogler, unpublished). In both cases, identification was based on phylogenetic analysis using PAUP 3.1.1 (Swofford, 1993) and the heuristic search option or test versions of PAUP 4 using the neighbour-joining option. If no ITS-RFLP match was found, we label the fungus by its family group name after replacing the ending with -oid followed by an RFLP pattern number (i.e. the first Amanitaceae unknown becomes

Amanitoid 1, the second Thelephoraceae unknown becomes Thelephoroid 2). Names reported here are the same as those used in Gardes & Bruns (1996).

Statistical analysis

We tested whether the fungal species associated equally with the two hosts. To adjust for the overwhelming abundance of Douglas-fir roots in most cores, we used the percentage by weight of ectomycorrhizas colonized by a fungus species out of the total ectomycorrhizas for each tree species in each core. These data deviated greatly from a normal distribution (percentages arcsine-transformed, Kolmogorov-Smirnov test for goodness of fit). We therefore used the Wilcoxon paired-sample test for this analysis; this is a non-parametric analogue to Student's *t*-test. The tests were conducted on fungi that occurred in six or more cores. It is unclear how powerful this test is when the sample size is low, although if all differences are of the same sign at $n = 6$, the significance level is $P < 0.05$ (Sokal & Rohlf, 1995). Two cores that contained only one tree host were not included for these analyses. We assume there was no intraspecific difference in host preference by the fungi. We considered cores as independent samples by assuming that the presence of a fungus in a neighbouring core does not influence host preference within a core.

RESULTS

No roots in the 25 cores had root hairs and all were considered to be colonized by ECM fungi. The biomass of ECM root tips per core was relatively evenly divided between the two tree species at the south end, and dominated by Douglas fir at the north end of the study site (Figs 1, 2). Two cores contained

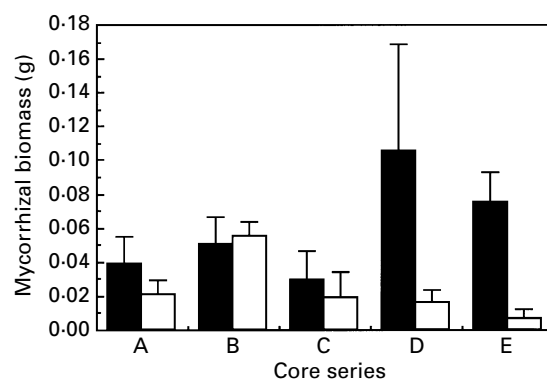


Figure 2. Biomass of ectomycorrhizas on Douglas fir (■) and bishop pine (□). Each core series shows the mean biomass $\pm$ 1 SE of five cores taken at each tree pair, A–E in Figure 1. Total soil volume sampled for each series is $c. 16 \times 10^3 \text{ cm}^3$. The ECM biomass was evenly distributed between the two tree species at core series A, B and C, but was primarily on Douglas fir at core series D and E. Two members of the Thelephoraceae made up most of the Douglas fir ECM biomass in core series D and E (data not shown).

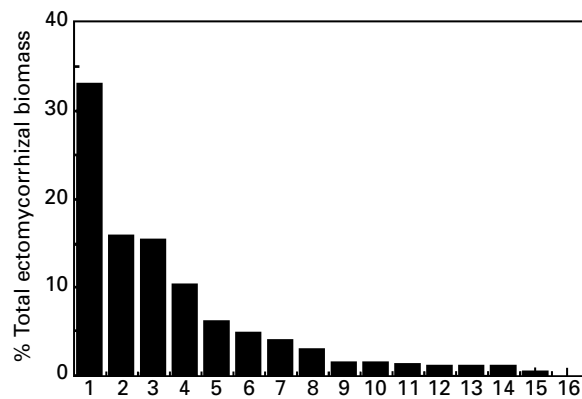


Figure 3. Percentage of total ECM biomass for each fungal species. 1, *Tomentella subulilacina*; 2, Thelephoroid 2; 3, *Russula amoenolens*; 4, *Russula xerampelina s.l.*; 5, *Rhizopogon parksii*; 6, *Amanita gemmata/pantherina*; 7, Cortinarioid 1; 8, *Cenococcum geophilum*; 9, *Amanita muscaria*; 10, unknown Ascomycete; 11, *Laccaria amethysteo-occidentalis*; 12, Amanitoid 1; 13, Thelephoroid 3; 14, *Xerocomus chrysenteron*; 15, Thelephoroid 4; 16, Russuloid 2.

roots of only one tree species; A5 had only bishop-pine roots and C5 had only Douglas-fir roots (Fig. 1). Seven other cores had relatively small amounts of bishop-pine roots (C1, C3, D3, D5, E1, E3, E4, E5; Fig. 1). Individual cores contained 3.8 ± 0.24 (mean ± 1 SE), fungal species per core with a range of one to six. One, one, nine, seven, five and two of the 25 cores had one, two, three, four, five and six fungal species, respectively.

Table 2. Species of fungi fruiting in the area and forming ectomycorrhizal root tips

Sporocarp	Collection number
<i>Amanita gemmata</i> (Fr.) Gill	trh 262
<i>Amanita muscaria</i> (L. ex Fr.) Hook.	tdb 1513
<i>Laccaria amethysteo-occidentalis</i> Mueller	trh 250
<i>Rhizopogon parksii</i> Smith	trh 125
<i>Russula amoenolens</i> Romagnesi	trh 300
<i>Russula xerampelina s.l.</i> (unidentified)	trh 216
<i>Tomentella subulilacina</i>	tdb 2015
<i>Xerocomus chrysenteron</i>	tdb 2029

Voucher specimens of the species are deposited in the herbarium at U.C. Berkeley.

In total, 16 fungal species were recorded on the roots (Fig. 3). This number includes seven RFLP/morphological types for which no matching sporocarps were found (Table 1, 2). The majority of fungal species were either infrequently encountered (Table 3), quantitatively minor (low biomass) (Fig. 3), or both. Four species in the Thelephoraceae and Russulaceae were among the most frequently encountered fungi and accounted for 74 % of the ECM biomass sampled (Fig. 3).

Five cores had two species in the Thelephoraceae (*Tomentella subulilacina* (Ellis & Holw.) Wakef. and Thelephoroid 2) and six cores had two species in the Russulaceae (*Russula amoenolens* Romagnesi and *R. xerampelina s.l.*). One core had three species in the Thelephoraceae, (*T. subulilacina*, Thelephoroid 2 and

Table 1. Restriction fragment sizes (base pairs) of ectomycorrhizal fungi and plants

Samples	<i>Alu</i> I	<i>Hinf</i> I	<i>Mbo</i> I
<i>Tomentella subulilacina</i> *†	458/(145)/111/86	366/223/126/107	374/227/196
Thelephoroid 2	454/144/86	373/129	256/226/195/119
Thelephoroid 3	459/238/87	377/185/127	227/196/143
Thelephoroid 4	460/145/90	374/225/153/128	375/226/195
<i>Russula amoenolens</i>	335/253/191/90	385/124	321/252/205
<i>Russula xerampelina s.l.</i>	504/(371)/283/90	401/353/124	334/216/169/99
Russuloid 2	489/255/93	397/371/123	577/257
<i>Amanita gemmata/pantherina</i> ‡	405/254/106/93	382/238/131	354/236
<i>Amanita muscaria</i>	419/262/110/93	435/118	353/244
Amanitoid 1	491/(236)/150/92	329/122	543/204
<i>Xerocomus chrysenteron</i>	542/184/157/95	431/245/179/40	361/298/154
<i>Laccaria amethysteo-occidentalis</i>	364/(288)/145/91	379/353/120	547/131/109
<i>Rhizopogon parksii</i>	718/95	245/128/113/75	310/256/233
Cortinarioid 1	554/250/156/96	225/215/198/129	573/280
<i>Cenococcum geophilum</i> I§	387/192	160/119/92	318/152
<i>Cenococcum geophilum</i> II§	715/391	591/158/108/90/63	392/312/145/122
Unknown Ascomycete§	735	255/131	412/238/63
<i>Pseudotsuga menziesii</i> ¶			296/270/190
<i>Pinus muricata</i> ¶			464/269

* RFLP patterns using ITS1f and ITS4b as primers unless otherwise noted.

† Band sizes shown in parentheses are submolar and diagnostic.

‡ *Amanita gemmata* and *A. pantherina* cannot be differentiated by their ITS-RFLP patterns.

§ ITS-RFLP patterns using ITS1f and ITS4 as primers for PCR product.

|| Two ITS-RFLP patterns were observed in *C. geophilum* because of the presence of an intron (Rogers *et al.*, 1993) in some samples.

¶ RFLP patterns using 28KJ and TW14 as primers for product.

Table 3. Frequency of fungi on Douglas fir and bishop pine ($n = 25$)

Ectomycorrhizal fungus	Number of cores		
	On Douglas fir only	On bishop pine only	On both trees
<i>Tomentella subulilacina</i>	7	2	4
Thelephoroid 2	9	0	2
<i>Russula amoenolens</i>	5	2	9
<i>Russula xerampelina</i> s.l.	2	0	5
<i>Rhizopogon parksii</i>	11	0	0
<i>Amanita gemmata/pantherina</i>	0	0	2
Cortinarioid 1	0	2	0
<i>Cenococcum geophilum</i>	6	3	4
<i>Amanita muscaria</i>	0	2	1
Unknown Ascomycete	1	0	2
<i>Laccaria amethysteo-occidentalis</i>	0	1	0
Amanitoid 1	1	4	1
Thelephoroid 3	0	0	2
<i>Xerocomus chrysenteron</i>	3	0	1
Thelephoroid 4	1	0	1
Russuloid 2	0	1	0

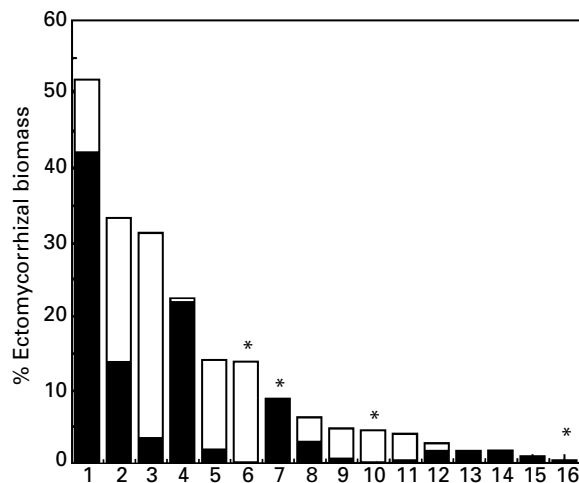


Figure 4. Percentage of ECM biomass each fungal species contributed to each tree's (□, bishop pine, ■, Douglas fir) total. Twelve of the sixteen fungi occurred on the roots of both tree species (see also Table 2). The four most abundant fungi belong to the Thelephoraceae and Russulaceae and contribute 74% of the total biomass of ectomycorrhizas sampled in the stand. Fungi that occurred on only one host are denoted with an *. 1, *Tomentella subulilacina*; 2, *Russula amoenolens*; 3, *Russula xerampelina* s.l.; 4, Thelephoroid 2; 5, *Amanita gemmata/pantherina*; 6, Cortinarioid 1; 7, *Rhizopogon parksii*; 8, *Cenococcum geophilum*; 9, *Amanita muscaria*; 10, *Laccaria amethysteo-occidentalis*; 11, Amanitoid 1; 12, unknown Ascomycete; 13, *Xerocomus chrysenteron*; 14, Thelephoroid 3; 15, Thelephoroid 4; 16, Russuloid 2.

Thelephoroid 3). The Thelephoraceae made up 46 % of the total Douglas fir ECM biomass in the study (Fig. 4), but some of this high biomass was caused by dense clusters of single species. A single core, D4, contributed 0.35 g of *T. subulilacina* ECM biomass to the Douglas-fir total; this was 51 % of the biomass collected for this type in the stand. Within core series E, a different species also in the

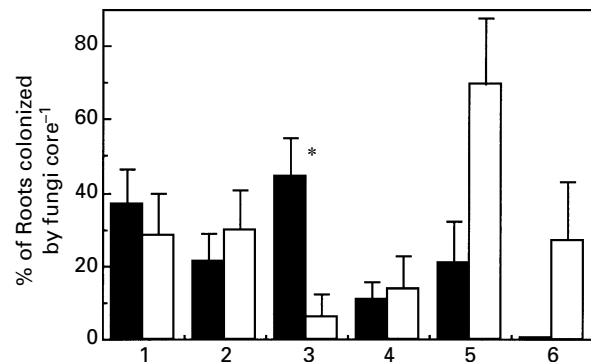


Figure 5. Mean (± 1 SE) colonization of tree roots (■, Douglas fir; □, bishop pine) by the most frequently occurring fungal species. Data are the percent of each tree's ectomycorrhizas colonized by the fungus in a core and represent only those cores where the fungus and both tree species were present. 1, *Tomentella subulilacina*; 2, *Russula amoenolens*; 3, Thelephoroid 2; 4, *Cenococcum geophilum*; 5, *Russula xerampelina* s.l.; 6, Amanitoid 1. Sample sizes are as follows: *Tomentella subulilacina*, $n = 12$; *Russula amoenolens*, $n = 14$; Thelephoroid 2, $n = 10$; *Cenococcum geophilum*, $n = 11$; *Russula xerampelina* s.l., $n = 7$; Amanitoid 1, $n = 6$. An asterisk indicates a significant difference between hosts in fungal colonization (Wilcoxon signed rank test, $P < 0.05$). Colonization of the two hosts was marginally different for *R. xerampelina* s.l. and Amanitoid 1 (Wilcoxon signed rank test, $0.10 > P > 0.05$).

Thelephoraceae (Thelephoroid 2) contributed 0.26 g dry biomass, 82 % of the total biomass for that fungus.

Of the 16 fungi, 12 were found on the root systems of both plants (Fig. 4, Table 3). All fungi that associated with both conifer species did so at least once within the same core and, in total, made up 90 % of the ECM biomass in the samples. Adjusting for the abundance of Douglas-fir ectomycorrhizas, there was no host preference shown by *T. subulilacina*, *R. amoenolens*, *C. geophilum* Fr. (Fig. 5). A preference

towards bishop pine might be indicated for *R. xerampelina* s.l. and Amanitoid 1, but in both cases, all differences were not of the same sign (see 'Materials and Methods'). Theleporoid 2 occurred on Douglas fir to a greater extent than on bishop pine. This is likely to be an artefact of the dominance of Douglas fir at the north end of the stand; Theleporoid 2 is a frequent and abundant associate of bishop pine in nearby stands (Gardes & Bruns, 1996).

Four of the 16 fungi were found associated with only one tree species, but three of these were too infrequent to assess their specificity (Fig. 4, Table 3). In total, these fungi contributed 10% of the ECM biomass in the stand. *Rhizopogon parksii* Smith was found in eleven cores and was specific to Douglas fir. Cortinarioid 1 occurred only on bishop pine in two cores which contained both tree species and was abundant in those cores, indicating that it was specific to bishop pine. *Laccaria amethysteae-occidentalis* Mueller occurred in a core that did not contain any Douglas-fir roots, but it commonly fruits in pure Douglas-fir forests in the area. Russuloid 1 occurred in only one core and in low abundance.

DISCUSSION

The aim of this study was to identify the ECM fungi on the root systems of Douglas fir and bishop pine in a mixed forest and to ascertain whether those fungi colonized one or both hosts. Before this study, fungal associations in a mixed ECM plant community were assessed by inference. Using a combination of ECM morphotyping and ITS-RFLP matching, we found that most species of fungi occurred frequently on the root systems of both plant species. Seven of the sixteen species were not matched to sporocarps but were identified as belonging to either the Ascomycota or various Basidiomycete families. Although we did not identify all ECM types to known species, we did find matching fungal ITS-RFLP patterns on both plant species and this gave an unambiguous demonstration that these fungi were not restricted to one host.

Most of the fungi associated with both plant species despite the overwhelming abundance of Douglas-fir roots in our samples (Figs 2, 4). All fungi that associated with both tree species did so in at least one 10 × 40 cm core (3141 cm³). This condition was observed with 12 of the 16 fungi in the stand, including the four most abundant species. These data support labelling experiments which strongly suggest that different plant species can be linked via a common mycelium (Björkman, 1960; Read, Francis & Finlay, 1985; Finlay & Read, 1986; Simard *et al.*, 1997a, b).

It is possible that different genotypes of a fungus might be specific to different tree species. We cannot

currently assess the population structure of these fungi from ectomycorrhizas alone. However, we predict that in most species, mycelium of an individual genet occupies a volume larger than our 10 × 40 cm cores. Results from clonal studies with *Suillus* and *Laccaria* spp. demonstrate that individuals are at least this large and can be as large as our entire stand (Baar, Ozinga & Kuyper, 1994; Dahlberg & Stenlid, 1994; De La Bastide, Kropp & Piché, 1994; Bonello, Bruns & Gardes, 1998). However, intermingling genets of a single species are still possible and we may have sampled more than one genet of a species within a core. Taxon specific primers are needed that will amplify DNA directly from ECM extracts and are useful for the identification of individual genotypes in a population (Egger, 1995).

Although this stand contained a mixture of Douglas fir and bishop pine, the pattern of species composition and ECM abundance was similar to that reported in Gardes & Bruns (1996) for a nearby pure stand of bishop pine. Half of the species and all four of the most common species occurred in both studies in about the same proportions (Fig. 3). The edaphic, climatic, and historical similarity and proximity of the two stands is likely to have played a larger role in fungal species-composition than host availability. This is consistent with the views of Molina *et al.* (1992) who state that Douglas fir and pines have an overlap of compatible fungi estimated at 72% or 1800 species.

The species of fungi present in this stand were also influenced by the disturbance regime inherent to bishop-pine forests. The fire periodicity in bishop-pine-communities is about once every 40 yr (Sugnet, 1985). Last, Dighton & Mason (1987) hypothesized that as a forest ages, changes in tree age and litter quality might cause a shift in the ECM community from wide-host-range fungi to narrow-host-range fungi. However, the disturbance periodicity of bishop pine stands is likely to be too short to allow such shifts in species composition to develop. Further, non-specific fungi might help to stabilize the forest community through disturbance events by associating with both pre-disturbance hosts and post-disturbance hosts (Perry *et al.*, 1989a). In this case, bishop-pine trees typically die in fires but rapidly re-establish through seed released from serotinous cones. Douglas fir must either survive a fire as a mature tree or recolonize the site from seed produced by local trees that did not die in the fire. Thus, fungi that can associate with both hosts have an advantage over those that do not, given the 40-yr fire regime of this community.

Theleporaceae and Russulaceae repeatedly emerge as abundant ectomycorrhizas in California plant communities (Gardes & Bruns, 1996; Bruns *et al.*, unpublished). Four species belonging to the Theleporaceae and Russulaceae made up most of

the ECM biomass in this study. We suspect that because these fungi colonized a great number of root tips, they probably explored a large volume of soil and contributed significantly to uptake of soil nutrients for both tree species. Further field efforts to determine actual fungal biomass and nutrient uptake in bulk soils is needed to ascertain these roles.

The only clearly host-specific fungus in the community was *Rhizopogon parksii*. This species has been reported to have an extremely narrow host-range, forming ectomycorrhizas exclusively with Douglas fir when grown in dual culture with other tree species (Massicotte *et al.*, 1994; Molina & Trappe 1994). Associating with a limited number of hosts might reduce carbon access for mycorrhizal fungi (Harley & Smith, 1983). Alternatively, specialists like *R. parksii* might be particularly efficient at acquiring photosynthate from their hosts and thereby gain greater access to carbon through specialization (Cullings *et al.*, 1996).

As a host-specific fungus, *R. parksii* might give Douglas fir exclusive access to resources and an avoidance of epiparasitism by other plant species via hyphal links (Molina *et al.*, 1992). However, *R. parksii* only accounted for 6% of the Douglas fir ECM biomass, an apparently minor component of its ECM root tips. The relative importance of this and other species might be underestimated because we sampled only the ectomycorrhizas. For instance, we did not sample the biomass of rhizomorphs by these fungi and *R. parksii* forms abundant rhizomorphs (Molina & Trappe, 1994) which function to transport water over ecologically important distances (Duddridge, Malibari & Read, 1980). The amount of hyphae in bulk soil of all species should be measured to further quantify their importance to plant hosts.

Our current knowledge of plant connections via hyphal linkages is based on a limited number of laboratory and even fewer field studies (Read *et al.*, 1985; Newman, 1988; Allen, 1991). Here we have provided a description of a mixed plant stand and its associated ECM community based on the biomass of ectomycorrhizas. Our results show that the majority of fungi in the stand occurred on both trees. The tree species may have been connected by mycelia. It is hard to argue otherwise considering the number of times a fungus occurred on both hosts within a single soil core. The potentially host-specific fungi amounted to only 10% of the ECM biomass (primarily *R. parksii* and Cortinarioid 1) whereas multiple-host-fungi amounted to 90% of the ECM biomass. The trees were colonized by the same ECM species thereby sharing similar capabilities for resource capture brought by the fungi to the trees. This suggests that competition by the trees for soil nutrients might, in part, be a function of their specific interactions with the common symbionts. Further studies are needed to demonstrate that individual fungal genets link plant species in the field

and to ascertain if such fungal individuals might provide differential nutrient translocation activities to different hosts (Finlay, 1989).

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